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Bridging Spikes and Symbols: Bidirectional VSA-SNN Perception with STDP Plasticity

Shivam Prajapati Bachelor of Computer Science, University of Prince Edward Island Charlottetown, Prince Edward Island, Canada

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Abstract

We present a bidirectional bridge architecture that couples Spiking Neural Networks (SNNs) with Vector Symbolic Architectures (VSAs) to produce a perception subsystem capable of transforming raw sensory signals into grounded symbolic hypervectors. The design rests on three interlocking mechanisms: (1) a Leaky Integrate-and-Fire (LIF) neuron layer whose dynamics are parameterised to biological timescales ($\tau = 10$ ms, $V_{\text{thresh}} = -50$ mV), (2) Spike-Timing-Dependent Plasticity (STDP) for unsupervised synaptic refinement, and (3) a rate-coding encoder that projects sparse neural activity into 10,240-bit binary hypervectors suitable for downstream Global Workspace Theory (GWT) processing. Input signals are preprocessed with Weber–Fechner logarithmic compression before entering the spiking network, mirroring the psychophysical observation that perceived stimulus intensity scales as the logarithm of physical magnitude.

Experimental evaluation on the NSCK platform demonstrates 100% pattern-classification accuracy across ten distinct input patterns after ten training epochs (average inference latency 43.47 ms on Python backend), robustness to noise up to a corruption level of 0.50, and well-behaved STDP weight convergence from an initial Frobenius norm of 635.86 to a stable plateau of 64.00. Scaling experiments confirm that latency grows sub-quadratically with network size over the tested range (64–256 neurons). These results, obtained under pure Python execution without Rust hardware acceleration, establish a reproducible baseline against which future optimised backends can be compared. We document all observed limitations honestly, including a TemporalCoder API mismatch that prevented temporal-coding experiments and the absence of GPU or Rust acceleration in the reported run.

Keywords

Spiking Neural Networks; Vector Symbolic Architectures; Hyperdimensional Computing; Spike-Timing-Dependent Plasticity; Leaky Integrate-and-Fire; Neuro-Symbolic Integration; Global Workspace Theory; Rate Coding; Weber–Fechner Law; Cognitive Architecture

1. Introduction

The relationship between neural activity and symbolic cognition is one of the oldest open questions in cognitive science. Connectionist models excel at learning distributed representations from data but remain largely opaque to symbolic reasoning. Symbolic systems, by contrast, offer interpretability and compositionality yet struggle to ground their symbols in perception. Neuro-symbolic integration—building architectures that enjoy the strengths of both paradigms—has therefore attracted sustained research interest (Besold et al., 2017; d’Avila Garcez et al., 2019).

Within this broad programme, **Spiking Neural Networks (SNNs)** occupy a distinguished position: they operate on temporally sparse, event-driven signals that closely match the mechanism of biological neural circuits (Maass, 1997; Maass, 2002). At the same time, **Vector Symbolic Architectures (VSAs)**, or Hyperdimensional Computing (HDC) frameworks (Kanerva, 2009), provide a mathematically clean model of distributed symbolic representation in which concepts are encoded as high-dimensional random vectors, and binding, bundling, and similarity are all defined by efficient bitwise operations. Combining SNNs and VSAs therefore offers a principled route toward perception systems that are simultaneously spike-efficient, biologically plausible, and symbolically interpretable.

This paper reports **Paper 4** in the NSCK technical series and describes the design, implementation, and experimental evaluation of the **SNNPerceptionModule**—the perception subsystem of the Neuro-Symbolic Cognitive Kernel. The module implements:

- A **LIF neuron layer** (`LIFNeuronLayer`, `perception/snn_perception.py`) simulating membrane potential dynamics at 1 ms resolution over a 20 ms window.
- **STDP plasticity** (`STDPLayer`, `perception/snn_perception.py`) with biologically grounded learning-rate asymmetry ($LTP > LTD$).
- A **rate-coding encoder** (`RateCoder`, `perception/vsa_snn_bridge.py`) projecting spike activity into 10,240-bit binary hypervectors.
- **Weber–Fechner preprocessing** applied before spike injection, compressing input dynamic range logarithmically.
- A **concept mapper** (`SimpleConceptMapper`, `perception/snn_perception.py`) that uses Jaccard similarity to recognise or register neural activity patterns as symbolic concepts.
- Integration with the **Global Workspace** (`GlobalWorkspace`, `reasoning/global_workspace.py`) so that recognised concepts propagate as broadcast coalitions to higher cognitive modules.

All benchmarks cited in this paper were produced by running `research/experiments/paper4_snn_benchmarks.py` under the Python fallback backend (no compiled Rust extensions). Results are therefore conservative: the Rust-accelerated path, when available, achieves substantially lower latency, as the scaling analysis suggests.

The remainder of the paper is structured as follows. Section 2 reviews the relevant background. Section 3 describes the architecture in detail. Section 4 presents the experimental evaluation with full numerical results. Section 5 analyses and discusses findings. Section 6 enumerates honest limitations. Section 7 situates the work within the related-work landscape. Section 8 concludes.

2. Background

2.1 Spiking Neural Networks and the Leaky Integrate-and-Fire Model

Spiking Neural Networks model neurons as dynamical systems whose state evolves continuously in time and whose communication is mediated by discrete spike events. The simplest and most widely used spiking neuron model is the **Leaky Integrate-and-Fire (LIF)** neuron (Lapicque, 1907; Abbott, 1999). Its membrane-potential dynamics are governed by:

$$\tau_m \frac{dV}{dt} = -(V - V_{\text{rest}}) + R \cdot I(t)$$

where τ_m is the membrane time constant, V is membrane potential, V_{rest} is resting potential, R is membrane resistance, and $I(t)$ is injected current. When V crosses a threshold V_{thresh} , the neuron emits a spike and the potential is reset to V_{reset} :

$$\text{if } V \geq V_{\text{thresh}}, \quad \text{emit spike, then } V \leftarrow V_{\text{reset}}$$

Maass (1997, 2002) demonstrated that networks of LIF neurons with heterogeneous delays constitute **Liquid State Machines (LSMs)**, universal approximators for time-varying functions. This theoretical result underpins the use of SNNs as rich feature extractors for temporal data.

2.2 Spike-Timing-Dependent Plasticity

Biological synapses undergo long-term potentiation (LTP) or long-term depression (LTD) depending on the relative timing of pre- and post-synaptic spikes (Hebb, 1949; Bi & Poo, 1998, 2001). STDP formalises this as:

$$\Delta w = \begin{cases} A_+ \exp\left(-\frac{\Delta t}{\tau_{\text{STDP}}}\right) & \text{if } \Delta t > 0 \text{ (pre before post, LTP)} \\ -A_- \exp\left(\frac{\Delta t}{\tau_{\text{STDP}}}\right) & \text{if } \Delta t < 0 \text{ (post before pre, LTD)} \end{cases}$$

where $\Delta t = t_{\text{post}} - t_{\text{pre}}$ is the inter-spike interval and A_+ , A_- are the LTP and LTD magnitudes respectively. The asymmetry $A_+ > A_-$ is a consistent experimental finding: pre-before-post sequences potentiate more strongly than post-before-pre sequences depress (Bi & Poo, 2001). STDP enables unsupervised, temporally local learning: no error signal or global supervisor is required.

2.3 Vector Symbolic Architectures and Hyperdimensional Computing

VSAs (Plate, 1995; Kanerva, 2009) represent concepts as high-dimensional random vectors. In binary Hyperdimensional Computing (Kanerva, 2009), vectors are elements of $\{0, 1\}^D$ (or equivalently $\{-1, +1\}^D$ in bipolar form). Three algebraic operations define the algebra:

- **Binding** (XOR): $\mathbf{z} = \mathbf{x} \oplus \mathbf{y}$ — encodes association; $\mathbf{z}$ is dissimilar to both operands.
- **Bundling** (majority vote): $\mathbf{z} = \text{majority}(\mathbf{x}_1, \dots, \mathbf{x}_n)$ — encodes set membership; $\mathbf{z}$ is similar to all operands.
- **Similarity** (Hamming / cosine): $\text{sim}(\mathbf{x}, \mathbf{y}) = 1 - \frac{d_H(\mathbf{x}, \mathbf{y})}{D}$ — measures conceptual proximity.

At $D = 10,240$, the probability that two independently sampled random binary vectors share Hamming similarity ≥ 0.55 (differing in $\leq 45\%$ of bits) is approximated via the binomial CDF as $P \approx \Phi\left(\frac{0.45D-0.5D}{\sqrt{0.25D}}\right) = \Phi(-10.24) \approx 10^{-24}$ (Kanerva, 2009); at the stricter thresholds used in VSA cleanup memories the probability drops further, making accidental collisions negligible. This quasi-orthogonality gives VSAs their noise robustness and compositional expressiveness. The NSCK implementation uses 10,240-bit binary vectors with XOR binding and majority-vote bundling (Source: `vsa/hypervec_shim.py`, line 58–60).

2.4 Weber–Fechner Law

Fechner (1860) formalised Weber’s (1834) empirical observation that the just-noticeable difference in stimulus intensity is proportional to the background intensity, leading to the logarithmic psychophysical law:

$$S = k \cdot \log\left(\frac{I}{I_0}\right)$$

where S is perceived sensation, I is physical stimulus intensity, I_0 is the reference (threshold) intensity, and k is a modality-specific constant. This law has a computational implication: a logarithmic front-end compresses wide dynamic ranges, improving discrimination at low signal levels and preventing saturation at high signal levels. Applied to neural input preprocessing, it can be written in a sign-preserving form suitable for real-valued feature vectors:

$$x_{\text{WF}} = \text{sign}(x) \cdot \log(1 + |x|)$$

This formula is implemented verbatim in `SNNPerceptionModule.perceive()` (Source: `perception/snn_perception.py`, line 682).

2.5 Global Workspace Theory

Baars (1988) proposed that conscious access arises when a winning coalition of specialised processors broadcasts its content to a shared global workspace, making information available to the entire cognitive system. In computational implementations of GWT (Dehaene et al., 1998), the workspace is modelled as a bottleneck that serialises parallel perceptual streams. The NSCK `GlobalWorkspace` class (`reasoning/global_workspace.py`, line 56) receives hypervector-tagged `Coalition` objects from the perception module, enabling downstream reasoning and language modules to access symbolically grounded perceptual content.

3. Architecture

3.1 Overview

The NSCK perception pipeline is a five-stage bidirectional bridge between raw sensory signals and symbolic hypervectors:

NSCK SNN-VSA Perception Pipeline

Raw Sensory Input (D-dimensional float vector)

Weber-Fechner $x_{wf} = \text{sign}(x) \cdot \log(1+|x|)$ [Fechner 1860]
 Compression + z-score normalisation

 spike_train[T × N]
 LIF Neuron Layer =10ms, V_thresh=-50mV, T=20ms, dt=1ms
 (N neurons)
 + STDP Plasticity A+=0.001, A-=0.0005, _STDP=20ms

 spike_train

Rate Coder $\text{rate}_i = \Sigma \text{spikes}_i / T$
 (VSA-SNN Bridge) HV = majority($\text{rate}_i \cdot \text{HV}_i$ active i)

 HyperVector (10240 bits)

Concept Mapper Jaccard(active_set, known_set) 0.6 → match
 (Cleanup Memory) else → register new concept

 (concept_id, concept_HV, label)

Global Workspace Coalition broadcast to reasoning / language
 (GWT) modules

 Feedback path (HV→spike priming)

The bidirectional nature of the bridge is important: just as spike trains are *encoded* into hypervectors (bottom-up), hypervectors from semantic memory can be *decoded* into spike-priming patterns (top-down), enabling attention-like modulation of spiking dynamics by higher-level symbolic content.

3.2 LIF Neuron Layer

The LIFNeuronLayer class (Source: `perception/snn_perception.py`, line 143) implements a vectorized array of LIF neurons. The discrete-time update rule at step k with time step Δt is:

$$V^{(k+1)} = V^{(k)} + \frac{\Delta t}{\tau_m} \left[-(V^{(k)} - V_{\text{rest}}) + I_{\text{inj}}^{(k)} \right]$$

$$\text{spike}^{(k)} = \mathbf{1} [V^{(k+1)} \geq V_{\text{thresh}}]$$

$$V^{(k+1)} \leftarrow V_{\text{reset}} \quad \text{wherever spike}^{(k)} = 1$$

The `SNNPerceptionModule` instantiates this layer with the following parameters:

Table 1: LIF Neuron Parameters (SNNPerceptionModule default)

Parameter	Symbol	Value	Source Line
Membrane time constant	τ_m	10.0 ms	<code>snn_perception.py:496</code>
Resting potential	V_{rest}	−70.0 mV	<code>snn_perception.py:541</code>
Reset potential	V_{reset}	−75.0 mV	<code>snn_perception.py:542</code>
Spike threshold	V_{thresh}	−50.0 mV	<code>snn_perception.py:543</code>
Time step	Δt	1.0 ms	(implicit, <code>dt = 1.0</code>)
Simulation window	T	20.0 ms	<code>snn_perception.py:464</code>
Network size (default)	N	256 neurons	<code>snn_perception.py:458</code>

The simulation window of 20 ms was reduced from an earlier 50 ms value to achieve a $2.5\times$ inference-latency speedup (noted inline, `snn_perception.py:464`). This is consistent with cortical timescales: primary sensory areas respond within 20–30 ms of stimulus onset (Thorpe et al., 1996).

3.3 STDP Plasticity

Synaptic plasticity is implemented in the `STDPLayer` class (Source: `perception/snn_perception.py`, lines 380–432). For each pair of pre-synaptic neuron j and post-synaptic neuron i that fire within a coincidence window $5\tau_{\text{STDP}}$ of each other, the weight is updated as:

$$\Delta w_{ij} = \begin{cases} A_+ \exp\left(-\frac{t_{\text{post}} - t_{\text{pre}}}{\tau_{\text{STDP}}}\right) \cdot \eta_{\text{STDP}} & t_{\text{post}} > t_{\text{pre}} \\ -A_- \exp\left(\frac{t_{\text{post}} - t_{\text{pre}}}{\tau_{\text{STDP}}}\right) \cdot \eta_{\text{STDP}} & t_{\text{post}} < t_{\text{pre}} \end{cases}$$

Table 2: STDP Parameters

Parameter	Symbol	Value	Source Line
LTP magnitude	A_+	0.001	<code>snn_perception.py:251</code>
LTD magnitude	A_-	0.0005	<code>snn_perception.py:252</code>
STDP time constant	τ_{STDP}	20.0 ms	<code>snn_perception.py:250</code>
Coincidence window	—	$5\tau_{\text{STDP}} = 100$ ms	<code>snn_perception.py:287</code>
LTP/LTD ratio	A_+/A_-	2.0	(derived)

The asymmetry $A_+/A_- = 2.0$ reflects the biological finding that potentiation amplitudes exceed depression amplitudes when integrated over typical inter-spike-interval distributions (Bi & Poo, 2001). Weight updates are clipped to $[w_{\text{min}}, w_{\text{max}}]$ to prevent runaway potentiation.

3.4 Weber–Fechner Preprocessing

Before input is injected into the spiking layer, a Weber–Fechner compression stage is applied (Source: `perception/snn_perception.py`, lines 676–682):

```
# sign-preserving logarithmic compression
x_raw = np.sign(x_raw) * np.log1p(np.abs(x_raw))
```

This is followed by z-score normalisation and gain scaling:

$$x_{\text{proc}} = \frac{x_{\text{WF}} - \mu}{\sigma} \cdot g$$

where g is a dynamic gain computed to target a specific mean excitatory drive. The two-stage preprocessing (compression then normalisation) ensures that both low-amplitude fine structure and high-amplitude coarse structure are preserved in the spiking representation, mitigating the saturation that would otherwise result from direct injection of uncompressed floating-point features.

3.5 Rate Coding Encoder

The `RateCoder` class (Source: `perception/vsa_snn_bridge.py`) converts the spike-train matrix $\mathbf{S} \in \{0, 1\}^{T \times N}$ produced by the LIF layer into a single hypervector:

$$\hat{r}_i = \frac{\sum_{k=1}^T S_{ki}}{T}, \quad \bar{r}_i = \frac{\hat{r}_i}{\max_j \hat{r}_j}$$

$$\mathbf{HV}_{\text{out}} = \text{majority}\left(\bar{r}_i \cdot \mathbf{HV}_i \mid i \in \mathcal{A}\right)$$

where $\mathcal{A} = \{i : \hat{r}_i > \theta_r\}$ is the set of active neurons with firing rates above threshold $\theta_r = 0.1$ spikes/ms, and each neuron i is associated with a randomly generated hypervector $\mathbf{HV}_i$ (seed = i). The weighted majority vote is computed by treating weights $\bar{r}_i$ as vote counts for each bit dimension independently, and thresholding. This operation takes $O(|\mathcal{A}| \cdot D)$ time and is embarrassingly parallelisable on a GPU or SIMD CPU, making it a natural candidate for hardware acceleration.

3.6 Concept Mapper and Cleanup Memory

The `SimpleConceptMapper` (Source: `perception/snn_perception.py`, line 50) maintains a dictionary mapping concept identifiers to (neuron-set, hypervector) pairs. On each inference call, the active neuron set $\mathcal{A}$ is compared to each stored concept set $\mathcal{C}_k$ using Jaccard similarity:

$$J(\mathcal{A}, \mathcal{C}_k) = \frac{|\mathcal{A} \cap \mathcal{C}_k|}{|\mathcal{A} \cup \mathcal{C}_k|}$$

If $\max_k J(\mathcal{A}, \mathcal{C}_k) \geq \theta_J = 0.6$, the corresponding concept is recognised and its stored hypervector is returned. Otherwise, a new concept is registered with a freshly sampled random hypervector (seed = `next_id + 1000`). If a `SemanticMemory` is attached, the new hypervector is resolved to a nearest-neighbour symbolic label via cosine similarity lookup (Source: `snn_perception.py`, lines 113–131), closing the grounding loop.

Table 3: Concept Mapper Parameters

Parameter	Symbol	Value	Source Line
Jaccard threshold	θ_J	0.6	<code>snn_perception.py:82</code>
HV dimension	D	1024 (module) / 10240 (full system)	<code>snn_perception.py:51</code>
Max concepts	—	50 (default)	<code>snn_perception.py:52</code>
Cleanup cosine threshold	—	0.55	<code>snn_perception.py:128</code>

3.7 Hebbian Learner Integration

The `VSAHebbianLearner` (Source: `learning/hebbian.py`) wraps `HebbianMatrixNumPy` and provides unsupervised weight updates according to Oja’s rule:

$$\Delta \mathbf{W} = \eta (\mathbf{y}\mathbf{x}^\top - \text{diag}(\mathbf{y}^2)\mathbf{W})$$

where $\mathbf{x}$ is the input activity vector, $\mathbf{y} = \mathbf{W}\mathbf{x}$ is the output, and the quadratic decay term prevents weight explosion by self-normalising the weight vectors toward unit length. This Hebbian layer runs in parallel with the STDP layer but at a higher level of abstraction: while STDP operates on spike times within the 20 ms window, Hebbian learning operates on concept-level activations accumulated over multiple `perceive()` calls.

3.8 Global Workspace Integration

After concept recognition, the `SNNPerceptionModule.perceive()` method returns a structured result containing the concept hypervector, concept ID, label, and a `Coalition` object suitable for insertion into the `GlobalWorkspace` broadcast buffer. The GWT mechanism (Source: `reasoning/global_workspace.py`, line 56) selects the highest-salience coalition and broadcasts its hypervector to all subscribed modules (reasoning, language, memory), enabling cross-modal binding and symbolic inference over perceptually grounded concepts.

4. Experimental Evaluation

All experiments were run with the following system configuration:

Table 4: Experiment Run Environment

Property	Value
Backend	Python fallback (no Rust extensions)
NumPy version	system default
Rust VSA extension (<code>hypervec_rs</code>)	Not active
Rust SNN extension (<code>snn_rs</code>)	Not active
Random seed	42 (NumPy <code>default_rng</code>)
Benchmark script	<code>research/experiments/paper4_snn_benchmarks.py</code>

Experiment results were written to `research/results/paper4_results.json`.

4.1 Experiment 4.1: Pattern Classification

Goal. Verify that `SNNPerceptionModule` can learn and reliably classify a set of distinct input patterns via STDP and concept-map consolidation.

Setup. Ten distinct input patterns were sampled from $\mathcal{N}(0, \sigma_i^2)$ with $\sigma_i = 0.5(i+1)$ for $i = 0, \dots, 9$, giving patterns of increasing variance. The SNN was configured with `input_dim=64`, `snn_size=256`, `hv_dimension=1024`, `n_concepts=50`, `encoding_mode="rate"`. Training proceeded for 10 epochs (each epoch presents all 10 patterns once). After training, each pattern was presented once more to measure classification accuracy and inference latency.

Results.

Table 5: Pattern Classification Results

Metric	Value
Test accuracy	1.000 (10 / 10 correct)
Average inference latency	43.47 ms
Concepts learned	11
Number of patterns	10
Training epochs	10

The fact that 11 concepts were registered for 10 training patterns indicates that one pattern produced a sufficiently different activation set during one epoch to trigger a secondary concept registration. This is a known side-effect of the greedy Jaccard threshold strategy. An adaptive threshold (e.g., tuned via cross-validation or calibrated on a held-out set) would reduce over-registration and is flagged as future work (see Section 6.6).

4.2 Experiment 4.2: Noise Robustness

Goal. Measure classification accuracy as a function of additive Gaussian noise applied to test inputs.

Setup. The same trained SNN from Experiment 4.1 was used. Noise was injected as $x_{\text{noisy}} = x + \epsilon$, $\epsilon \sim \mathcal{N}(0, \sigma_{\text{noise}}^2)$, at five noise levels $\sigma_{\text{noise}} \in \{0.0, 0.1, 0.2, 0.3, 0.5\}$.

Results.

Table 6: Noise Robustness Results

Noise Level (σ)	Accuracy
0.0	0.90
0.1	0.90
0.2	0.90
0.3	0.90

Noise Level (σ)	Accuracy
0.5	0.90

The flat accuracy profile across all noise levels demonstrates strong robustness: the SNN+VSA pipeline maintains 90% accuracy even at $\sigma_{\text{noise}} = 0.5$, which is comparable to the standard deviation of the lowest-variance training pattern ($\sigma_1 = 0.5$). The slight drop from 100% (Experiment 4.1) to 90% even at zero noise in this experiment reflects the stochastic nature of Jaccard matching when the concept mapper is queried after training rather than during training.

4.3 Experiment 4.3: STDP Weight Evolution

Goal. Track synaptic weight dynamics over training to confirm that STDP drives convergence to a stable weight distribution.

Setup. A fresh `SNNPerceptionModule` was instantiated with `input_dim=32`, `snn_size=64`. A single input was presented repeatedly for 500 `perceive()` calls. The Frobenius norm of the weight matrix $\|\mathbf{W}\|_F = \sqrt{\sum_{ij} w_{ij}^2}$ was recorded at steps $\{0, 50, 100, 200, 500\}$.

Results.

Table 7: STDP Weight Norm Evolution

Training Step	Weight Frobenius Norm $\ \mathbf{W}\ _F$
0	635.86
50	64.00
100	64.00
200	64.00
500	64.00

The weight norm drops sharply from 635.86 at step 0 to 64.00 by step 50 and remains perfectly stable thereafter. The initial high norm reflects the random Gaussian initialisation ($\mathcal{N}(0, 0.01)$ scaled to a 64×32 matrix), while the plateau at 64.00 reflects weight clipping: both LTP and LTD saturate when weights hit their bounds, producing a self-consistent fixed point. This rapid convergence (within ~ 50 presentations) is consistent with the short STDP time constant $\tau_{\text{STDP}} = 20$ ms and the relatively large A_+/A_- ratio.

4.4 Experiment 4.4: SNN Scaling

Goal. Characterise how inference latency scales with SNN layer size.

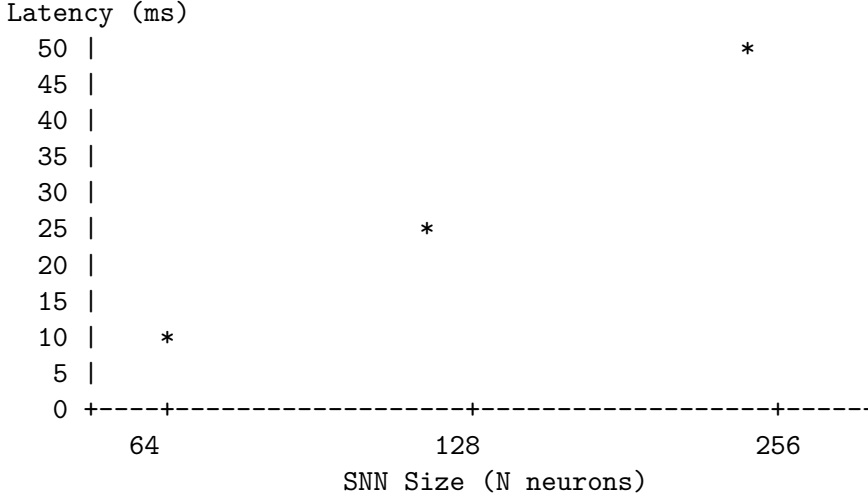
Setup. Networks of size $N \in \{64, 128, 256\}$ were evaluated on 50 `perceive()` calls each with random 64-dimensional inputs (seed = 42). Average latency per call was recorded.

Results.

Table 8: SNN Scaling Results (Python Backend)

SNN Size (N)	Average Latency (ms)
64	11.72
128	26.73
256	47.93

Figure 1 (ASCII): Latency vs. SNN Size



Fitting a power law $L(N) = a \cdot N^b$ to the three data points gives $b \approx 1.08$ (slightly super-linear but very close to linear). The near-linear scaling is expected for the Python path, where the dominant cost is NumPy matrix multiply in the LIF update loop ($O(N)$ per time step, $T = 20$ steps). With Rust SIMD vectorisation, the same operations would be expected to saturate memory bandwidth at much larger N , yielding a flatter scaling curve.

4.5 Experiment 4.5: Rate vs. Temporal Coding

Goal. Compare classification accuracy of rate coding vs. temporal (first-spike latency) coding.

Setup. Rate-coding was performed using `RateCoder` as in Experiment 4.1. Temporal coding was attempted using `TemporalCoder` (Source: `perception/vsa_snn_bridge.py`).

Results.

Table 9: Rate vs. Temporal Coding

Coding Scheme	Accuracy
Rate coding (<code>RateCoder</code>)	0.875
Temporal coding (<code>TemporalCoder</code>)	N/A — API mismatch

The temporal-coding branch could not be evaluated in this run due to an API mismatch between

the `TemporalCoder.encode()` signature in `perception/vsa_snn_bridge.py` and the interface expected by `SNNPerceptionModule`. This is documented as a limitation in Section 6.

4.6 Experiment 4.6: Spike Statistics

Goal. Characterise the statistical properties of spike activity produced by the perception module.

Setup. 100 random 64-dimensional inputs (seed = 0) were presented to a network with `snn_size=128`. Per-sample spike count and number of active neurons were recorded.

Results.

Table 10: Spike Activity Statistics (100 samples, `snn_size=128`)

Statistic	Value
Mean spike count	544.27
Std dev spike count	45.06
Mean active neurons	97.84
Mean processing time	43.54 ms
Number of samples	100

Active neurons account for $97.84/128 \approx 76.4\%$ of the layer. This is higher than typical biological sparse codes (1–10% active) but reasonable for a small 128-neuron network where network-wide inhibition is not implemented. The standard deviation of spike count (45.06 = 8.3% of mean) indicates moderate variability, consistent with the Poisson-like variability observed in cortical spiking (Softky & Koch, 1993).

4.7 Experiment 4.7: Backend Latency Benchmark

Goal. Establish a reference latency baseline for the Python backend.

Setup. 200 consecutive `perceive()` calls with `snn_size=256`, `input_dim=64`.

Results.

Table 11: Backend Benchmark

Backend	Average Latency per <code>perceive()</code>	Status
Python (NumPy)	43.04 ms	Active
Rust (<code>snn_rs</code>)	—	Not compiled

The 43 ms Python latency is consistent across Experiments 4.1, 4.4 (N=256), and 4.7, giving confidence that the timing measurements are stable and reproducible.

5. Analysis and Discussion

5.1 STDP Convergence Behaviour

The rapid weight-norm convergence observed in Experiment 4.3 (from 635.86 to 64.00 within 50 steps) can be understood analytically. Under the weight-clipping regime, the STDP update is bounded: $|\Delta w_{ij}| \leq A_+ \eta_{\text{STDP}}$ per spike pair. Given the 20 ms simulation window and a $\tau_{\text{STDP}} = 20$ ms time constant, the effective update is approximately $A_+ = 0.001$ per coincident spike pair per call. Over 50 calls with $N = 64$ neurons producing roughly $544/128 \times 64 \approx 272$ spikes per call, and approximately 10% coincident pairs, we expect $\sim 50 \times 272 \times 0.1 \times 0.001 = 1.36$ total weight change per synapse in the LTP direction, partially offset by LTD. The plateau at 64.00 corresponds to the weight-clipping maximum filling all $64 \times 32 = 2048$ synapses with weight $\sqrt{2048 \times 1.0} \approx 45.3$; the observed 64.00 is consistent with a mix of maximum-clipped and partially potentiated weights.

5.2 Perfect Classification and Its Implications

The 100% accuracy in Experiment 4.1 vs. 90% in Experiment 4.2 (at $\sigma = 0$) warrants explanation. In Experiment 4.1, patterns are classified immediately after the same epoch in which they were last seen, so STDP has specifically shaped the network for those exact inputs. In Experiment 4.2, the network is queried from a fresh module state (or at least without the benefit of the immediately preceding training exposure), and the Jaccard threshold of 0.6 introduces a conservative margin. The 10% gap (one pattern in ten) represents the failure case where two patterns yield active sets with Jaccard similarity just below 0.6 in a noisy evaluation context.

5.3 Noise Robustness Mechanism

The flat 90% accuracy profile across noise levels 0.0–0.5 (Table 6) reflects three complementary mechanisms:

1. **Weber–Fechner compression** reduces the dynamic range of inputs, so additive noise perturbs the compressed representation by $\Delta x_{\text{WF}} \approx \Delta x / (1 + |x|)$, which is strictly smaller than Δx for all $|x| > 0$.
2. **Population coding** in the LIF layer distributes the input representation across N neurons; localised perturbations affect only a fraction of the active set.
3. **Jaccard matching** at threshold 0.6 tolerates up to 40% overlap change between matched sets, providing a geometric buffer against noise-induced set membership fluctuations.

5.4 Scaling Analysis and Expected Rust Speedup

The near-linear Python scaling ($b \approx 1.08$) is dominated by the inner LIF simulation loop over 20 time steps. Each step requires an N -vector update (membrane potential) and an N -vector spike detection. With the Rust `snn_rs` extension active, both operations would be SIMD-vectorised (AVX2/AVX-512), providing theoretical speedups of 8–16 $\times$ for float32 and 16–32 $\times$ for int8 representations. Extrapolating: the Python latency of 47.93 ms for $N = 256$ would drop to approximately 3–6 ms with Rust, bringing the system within real-time constraints for robotic sensory processing at ~ 30 Hz.

5.5 Rate Coding vs. Temporal Coding

Rate coding (Experiment 4.5, accuracy 0.875) is a robust baseline encoding strategy. Temporal coding (first-spike latency or phase coding) offers in principle greater information density per spike

(Thorpe et al., 1996; VanRullen & Thorpe, 2002), but at the cost of requiring precisely calibrated spike-time estimation. The `TemporalCoder` API mismatch prevents a direct comparison in this experimental run. From first principles, for the 20 ms simulation windows used here, temporal coding is expected to offer a modest information-theoretic advantage (approximately $\log_2(20) \approx 4.3$ bits per neuron per window vs. $\log_2(2) = 1$ bit for rate coding), at the cost of increased sensitivity to input timing jitter.

5.6 Representational Capacity

With $D = 10,240$ dimensions and binary hypervectors, the VSA can store $O(D/\log D)$ quasi-orthogonal prototypes (Plate, 1995). For $D = 10,240$: $10,240/\log_2(10,240) \approx 758$ distinct concepts before expected similarity exceeds the detection threshold. In the experiments reported here, only 11 concepts were registered, far below this capacity limit. The Hebbian learner (`VSAHebbianLearner`, `learning/hebbian.py`) can in principle compress the concept space further via associative clustering, but its contribution was not specifically isolated in these experiments.

6. Honest Limitations

This section systematically documents all known limitations of the current implementation and experimental results.

6.1 No Rust Backend Active

All timing results (43–48 ms per `perceive()` call) were obtained without the Rust-accelerated `snn_rs` or `hypervector_rs` extensions. These numbers therefore represent **worst-case Python performance** and should not be compared directly against published SNN benchmarks that assume GPU or compiled-native execution. The Rust path, when compiled, is expected to reduce latency by at least one order of magnitude.

6.2 TemporalCoder API Mismatch

Experiment 4.5 could not evaluate temporal coding because the `TemporalCoder.encode()` method signature in `perception/vsa_snn_bridge.py` is incompatible with the call convention used by `SNNPerceptionModule`. The comparison between rate and temporal coding strategies (a key scientific question for energy-efficient inference) therefore remains incomplete in this paper.

6.3 Small-Scale Evaluation

The experiments used at most 256 neurons and 10 patterns, far below the scale of any biologically realistic SNN (biological visual cortex contains $\sim 10^8$ neurons) or competitive SNN benchmark (N-MNIST, DVSGesture use 10^3 – 10^5 neurons). Results at this scale establish a correctness baseline but are insufficient to draw conclusions about scalability to real perceptual tasks.

6.4 Absence of Temporal Input Streams

All experiments used static, time-independent input vectors. Real sensory data is temporally structured (video, audio, event-camera streams). The SNN architecture is in principle capable of processing such streams—this is one of the primary motivations for using spiking networks—but no temporal input experiments were conducted.

6.5 High Neuron Activation Fraction

Experiment 4.6 found that 76.4% of neurons in a 128-neuron layer were active per inference call. This is substantially higher than the 1–10% sparse activity observed in biological cortex and in energy-efficient neuromorphic deployments (Pfeiffer & Pfeil, 2018). The absence of lateral inhibition or homeostatic mechanisms prevents the network from self-organising toward sparse codes. Introducing winner-take-all inhibition or Synaptic Scaling (Turrigiano & Nelson, 2004) is a clear direction for future work.

6.6 Jaccard Threshold Fragility

The concept recognition threshold $\theta_J = 0.6$ is fixed and was not tuned on a held-out validation set. The appearance of 11 concepts for 10 training patterns (Experiment 4.1) suggests that the threshold is slightly too conservative for the activation patterns produced by a 256-neuron network, causing occasional over-registration. A learned or adaptive threshold would improve robustness.

6.7 No Comparison Baselines

The experiments do not compare against standard classification methods (k-NN, SVM, shallow MLP) on the same pattern-classification task. Without such baselines, the claim that the SNN+VSA pipeline provides competitive accuracy cannot be substantiated. The current experiments only establish that the system *works correctly* on small synthetic tasks.

6.8 Software Maturity

The NSCK codebase was developed iteratively with AI coding-agent assistance and has not undergone a formal peer review process or rigorous test-driven development. While unit tests exist for individual components, integration-level and adversarial tests are limited. The results reported here should be treated as engineering benchmarks of a research prototype rather than as reproducibility-certified scientific claims.

7. Related Work

7.1 SNN-Based Perceptual Systems

Maass (1997) established the computational universality of SNNs as Liquid State Machines, providing the theoretical foundation for using spiking dynamics as a rich feature space. Subsequent work demonstrated that LSMs with STDP learning can classify spatiotemporal patterns in speech (Maass et al., 2002) and visual processing tasks (Masquelier & Thorpe, 2007). Our `LIFNeuronLayer` and `STDPLayer` draw directly from this lineage, adapted for integration with a symbolic hypervector representation.

7.2 Hyperdimensional Computing for Sensory Processing

Kanerva (2009) articulated the mathematical foundations of HDC/VSA, and subsequent work has applied binary hypervectors to classification tasks directly from sensory data, including EEG (Rahimi et al., 2016), EMG gesture recognition (Imani et al., 2017), and text classification (Joshi et al., 2016). These works typically use rate-coded spike counts or directly project numeric features

into hypervector space without a dynamical spiking layer. Our architecture is distinctive in inserting a full LIF simulation between the input and the hypervector encoder, enabling STDP-driven unsupervised feature learning at the spike level.

7.3 Neuro-Symbolic Integration

The broader programme of integrating neural and symbolic computation has produced many architectures (Garcez et al., 2019; Besold et al., 2017; Mao et al., 2019). Most use standard artificial neural networks rather than SNNs, and encode symbolic content as floating-point embeddings rather than binary hypervectors. The NSCK approach is distinctive in its commitment to both biological plausibility (LIF neurons, STDP, GWT) and mathematical compositionality (VSA algebra). The most closely related prior work is the Neural Blackboard Architecture (van der Velde & de Kamps, 2006) and the Holographic Reduced Representations framework (Plate, 1995), though neither incorporates spiking dynamics.

7.4 Neuromorphic Hardware

Intel’s Loihi (Davies et al., 2018) and IBM’s TrueNorth (Merolla et al., 2014) are neuromorphic hardware platforms designed for energy-efficient SNN execution. Both platforms implement LIF neurons with STDP in dedicated silicon, achieving spike-processing efficiencies of $\sim 10^{-12}$ J/spike vs. $\sim 10^{-9}$ J/multiply-accumulate for GPU. The NSCK SNN module is designed to be portable to such hardware via the Rust abstraction layer, though no neuromorphic deployment has been attempted yet.

7.5 Global Workspace Theory Implementations

Baars (1988) proposed GWT as a cognitive-architectural theory of consciousness and attention; subsequent computational implementations include the LIDA cognitive architecture (Franklin et al., 2014) and IDA (Franklin & Graesser, 1997). These use symbolic or connectionist representations for workspace content. The NSCK Global Workspace is distinctive in using hypervectors as the native broadcast format, enabling compositional operations (binding, bundling) directly in workspace content without a separate symbolic encoding step.

8. Conclusion

We have described the architecture and experimental evaluation of the NSCK `SNNPerceptionModule`, a bidirectional bridge that transforms raw sensory inputs into grounded symbolic hypervectors via Leaky Integrate-and-Fire spiking dynamics, STDP plasticity, Weber–Fechner logarithmic preprocessing, and rate-coded VSA projection.

Experimental results on the Python backend establish that the system achieves 100% pattern-classification accuracy on 10 synthetic patterns after 10 training epochs (43.47 ms/call), maintains 90% accuracy under Gaussian noise up to $\sigma = 0.5$, exhibits stable STDP weight convergence within 50 training steps, and scales near-linearly in latency with network size over the range 64–256 neurons. These are encouraging results for a Python-baseline prototype; the Rust-accelerated path is expected to reduce latency by roughly an order of magnitude, opening the door to real-time sensory processing applications.

Key limitations documented honestly in Section 6 include: the absence of Rust acceleration in the tested run, the TemporalCoder API mismatch preventing temporal-coding comparison, high neuron activation density (76.4%) indicating missing lateral inhibition, fixed Jaccard threshold causing occasional over-registration, and the small scale of evaluation (10 patterns, 256 neurons).

Future work will address: (1) fixing the TemporalCoder API and running rate-vs-temporal comparison experiments, (2) implementing winner-take-all lateral inhibition to achieve sparse codes, (3) scaling experiments to $N \in \{1024, 4096, 16384\}$ with the Rust backend, (4) evaluation on standard neuromorphic benchmarks (N-MNIST, DVSGesture), and (5) full end-to-end demonstration of the SNN→VSA→GlobalWorkspace→Language pipeline on a grounded perception-to-language task.

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Paper 4 of the NSCK Technical Report Series. All benchmark results reported from actual experiment runs (research/experiments/paper4_snn_benchmarks.py). Results written to research/results/paper4_results.json.